Errors & Exceptions

Error Trapping

Learning Outcome of This Session

1. What is an exception?
2. Error types during coding
3. Python Built-in Exceptions
4. Create custom exceptions
5. How to handle errors?

What is an Exception?

- Exception is an event which will occur during the execution of a program that will halt or disrupt the flow of the program execution.
- A programmer should handle these errors properly which will safeguard program execution, or it will allow the program to recover from the exception

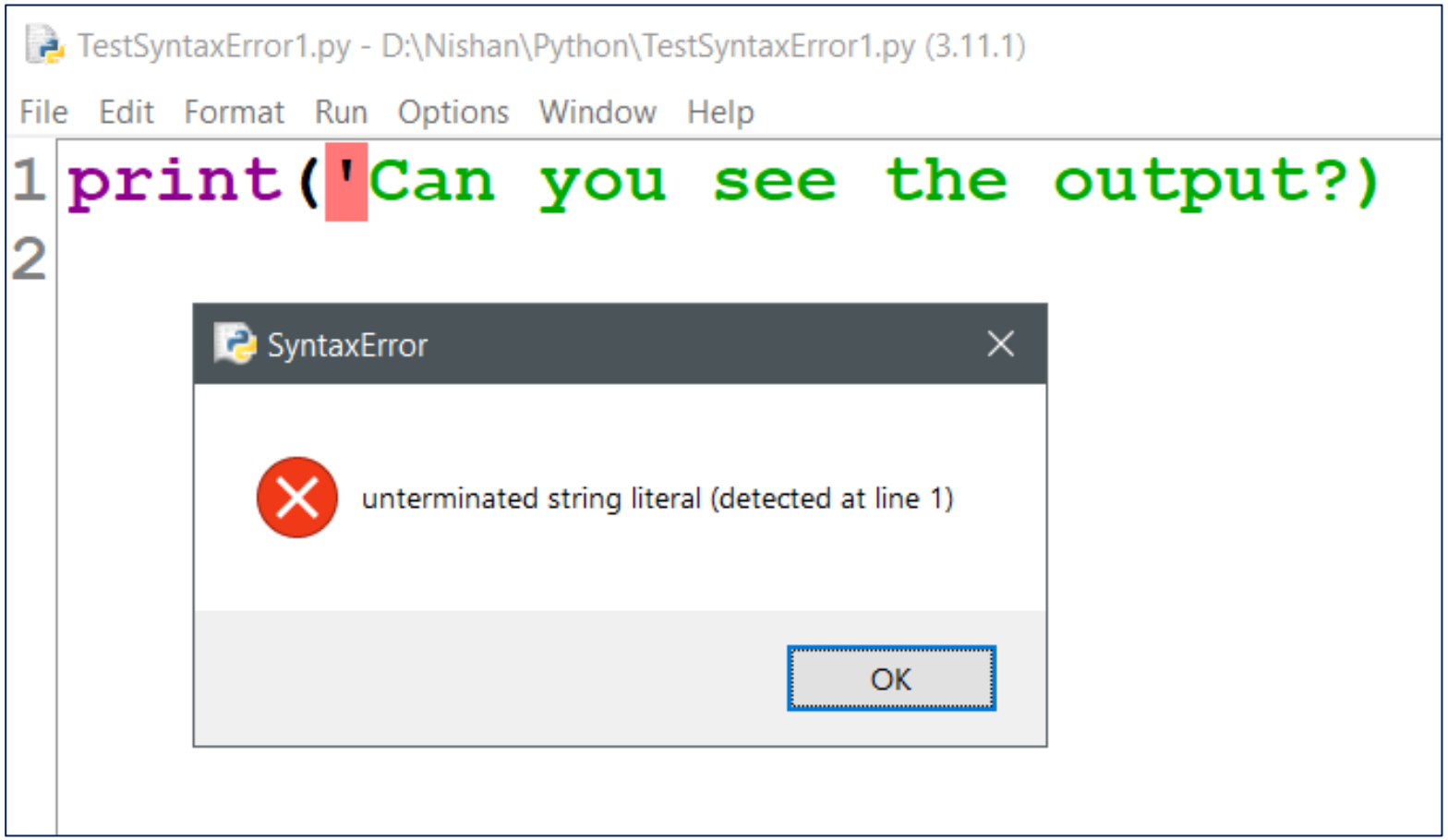
Error Types in Coding

- Syntax Errors
- Runtime Errors
- Logical Errors

Syntax Errors

- Easy to find
- Occurs during the coding, testing and development
- Easily fixed
- Program is released once the syntax errors are fixed

Syntax Error – Example 1



Syntax Error – Example 2

What are the details you see in this error message?

TestSyntaxError2.py - D:\Nishan\Python\TestSyntaxError2.py (3.11.1)

File Edit Format Run Options Window Help

```
1 message = "Hello"
2 while c<5:
3     print(message)
4     c += 1
5
```

```
========== RESTART: D:\Nishan\Python\TestSyntaxError2.py =====
Traceback (most recent call last):
  File "D:\Nishan\Python\TestSyntaxError2.py", line 2, in <module>
    while c<5:
NameError: name 'c' is not defined
>>>
```

Runtime Errors

- Moderately difficult to find
- Occurs during the deployment
- Once found, easy to fix

Runtime Error – Example

TestRuntimeError.py - D:\Nishan\Python\TestRuntimeError.py (3.11.1)

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```
1 fo = open("I:\\data.txt","w")
2 fo.write("hello")
3 fo.close()
4
```

```
========== RESTART: D:\Nishan\Python\TestRuntimeError.py ========
Traceback (most recent call last):
  File "D:\Nishan\Python\TestRuntimeError.py", line 1, in <module>
    fo = open("I:\\data.txt","w")
FileNotFoundError: [Errno 2] No such file or directory: 'I:\\data.txt'
>>>
```

```
========== RESTART: D:\Nishan\Python\TestRuntimeError.py ====
Traceback (most recent call last):
  File "D:\Nishan\Python\TestRuntimeError.py", line 1, in <module>
    fo = open("C:\\data.txt","w")
PermissionError: [Errno 13] Permission denied: 'C:\\data.txt'
>>>
```

Logical Errors

- Hard to find
- Algorithm errors
- May visible during deployment not during coding, testing and development
- Can be fixed if figured properly
- Program may be released with the logical error
- Shows no errors!
- Unpredicted behavior or output

Logical Error – Example

LogicalError.py - D:\Nishan\Python\LogicalError.py (3.11.1)

File Edit Format Run Options Window Help

```
1 val1 = input("Value 1 : ")
2 val2 = input("Value 2 : ")
3 print("Addition is ", val1 + val2)
4
```

```
========== RESTART: D:\Nishan\Python\LogicalError.py ==
Value 1 : 5
Value 2 : 6
Addition is  56
>>>
```

Exception Handling in Python

- Python uses 4 different keywords to handle errors
- We discuss 3 of them
- They are,
  - try
  - except
  - else

try … except

try



You ask the code to “try”
a segment. If it works, then no issues

try … except

try



The “try” throws an error if it ***fails to execute*** a code properly

The “except” captures the error. Programmer should handle it.



except

Example Exception – Case 1

Exception1.py - D:\Nishan\Python\Exception1.py (3.11.1)

File Edit Format Run Options Window Help

```
1 print(value)
2
```

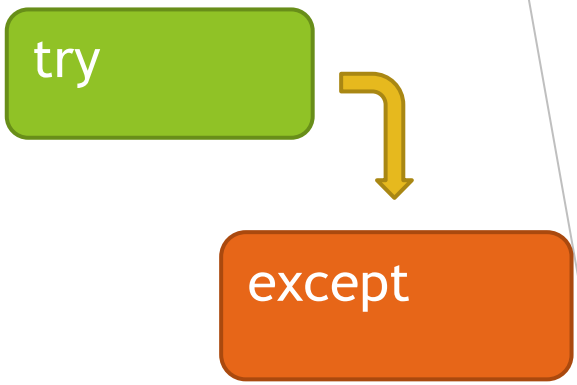
```
========== RESTART: D:\Nishan\Python\Exception1.py ====
Traceback (most recent call last):
  File "D:\Nishan\Python\Exception1.py", line 1, in <module>
    print(value)
NameError: name 'value' is not defined. Did you mean: 'False'?
>>>
```

Example Exception Handling – Case 1

Exception1Handle.py - D:\Nishan\Python\Exception1Handle.py (3.11.1)

File Edit Format Run Options Window Help

```
1 try:
2     print(value)
3 except:
4     print("Error found")
5
```




```
========== RESTART: D:\Nishan\Python\Exception1Handle.py ==
Error found
>>>
```

Capture a Specific Error / Multiple Errors

Specific Error

General Error

Exception2MultipleErrors.py - D:/Nishan/Python/Exception2MultipleErrors.py (3.11.1)

File Edit Format Run Options Window Help

```
1 try:
2     print(value)
3
4 except NameError:
5     print("Name error was captured")
6
7 except:
8     print("General capture")
9
```

```
========== RESTART: D:/Nishan/Python/Exception2MultipleErrors.py ==
Name error was captured
>>>
```

Python Exceptions (Built-In)

- Some common exceptions are,
  - NameError
  - ImportError
  - IndexError
  - SyntaxError
  - TypeError
- You can read more about these exceptions via https://docs.python.org/3/library/exceptions.html#exception-hierarchy

Exception hierarchy

The class hierarchy for built-in exceptions is:

```
BaseException
 ├── BaseExceptionGroup
 ├── GeneratorExit
 ├── KeyboardInterrupt
 ├── SystemExit
 └── Exception
      ├── ArithmeticError
      │    ├── FloatingPointError
      │    ├── OverflowError
      │    └── ZeroDivisionError
      ├── AssertionError
      ├── AttributeError
      ├── BufferError
      ├── EOFError
      ├── ExceptionGroup [BaseExceptionGroup]
      ├── ImportError
      │    └── ModuleNotFoundError
      ├── LookupError
      │    ├── IndexError
      │    └── KeyError
      ├── MemoryError
      ├── NameError
      │    └── UnboundLocalError
      ├── OSError
      │    ├── BlockingIOError
      │    ├── ChildProcessError
      │    ├── ConnectionError
      │    │    ├── BrokenPipeError
      │    │    ├── ConnectionAbortedError
      │    │    ├── ConnectionRefusedError
      │    │    └── ConnectionResetError
      │    │
```

Your Own Exceptions

- You can create your own errors
- Raise your exception

Raise Exception – Case 1

RaiseException1.py - D:\Nishan\Python\RaiseException1.py (3.11.1)

File Edit Format Run Options Window Help

```
1 uni = input("Type your university : ")
2 if uni != "IIT":
3     raise Exception("OMG You don't know the university?")
4 else:
5     print("Very Good")
6
```

```
========== RESTART: D:\Nishan\Python\RaiseException1.py ===
Type your university : IIT
Very Good
>>>
```

```
========== RESTART: D:\Nishan\Python\RaiseException1.py ===
Type your university : SLIIT
Traceback (most recent call last):
  File "D:\Nishan\Python\RaiseException1.py", line 3, in <module>
    raise Exception("OMG You don't know the university?")
Exception: OMG You don't know the university?
>>>
```

Raise Exception Handling – Case 1

RaiseException2Handle.py - D:\Nishan\Python\RaiseException2Handle.py (3.11.1)

File Edit Format Run Options Window Help

```
try:
    uni = input("Type your university : ")
    if uni != "IIT":
        raise Exception("OMG You don't know the university?")
    else:
        print("Very Good")

except:
    print("Kick this guy out")

```

```
========== RESTART: D:\Nishan\Python\RaiseException2Handle.py ==
Type your university : IIT
Very Good
>>>
```

```
========== RESTART: D:\Nishan\Python\RaiseException2Handle.py ==
Type your university : SLIIT
Kick this guy out
>>>
```

Let’s Try 2 Simple Programs

A Simple Program to See an Error

SimpleProgram1.py - D:\Nishan\Python\SimpleProgram1.py (3.11.1)

File Edit Format Run Options Window Help

```
1 friends = ['Smith','Jane','Nazeem','Judi','Don']
2 number = int(input("Type a number to find your friend : "))
3 print("Your friend is :",friends[number])
4
```

Program Output

```
========== RESTART: D:\Nishan\Python\SimpleProgram1.py ===
Type a number to find your friend : 2
Your friend is : Nazeem
>>>
========== RESTART: D:\Nishan\Python\SimpleProgram1.py ===
Type a number to find your friend : 1
Your friend is : Jane
>>>
========== RESTART: D:\Nishan\Python\SimpleProgram1.py ===
Type a number to find your friend : 6
Traceback (most recent call last):
  File "D:\Nishan\Python\SimpleProgram1.py", line 3, in <module>
    print("Your friend is :",friends[number])
IndexError: list index out of range
```

A Simple Program Which Handles The Error

SimpleProgram1Handles.py - D:/Nishan/Python/SimpleProgram1Handles.py (3.11.1)

File Edit Format Run Options Window Help

```
1  friends = ['Smith','Jane','Nazeem','Judi','Don']
2  try:
3      number = int(input("Type a number to find your friend : "))
4      print("Your friend is :",friends[number])
5
6  except IndexError:
7      print("No such friend name under that number")
8  else:
9      print("Friend found")
10
```

Program Output

```
============== RESTART: D:/Nishan/Python/SimpleProgram1Handles.py ===
Type a number to find your friend : 2
Your friend is : Nazeem
Friend found
>>>
============== RESTART: D:/Nishan/Python/SimpleProgram1Handles.py ===
Type a number to find your friend : 8
No such friend name under that number
>>>
```

A Simple Program to Raise an Error

SimpleProgram2Raise.py - D:\Nishan\Python\SimpleProgram2Raise.py (3.11.1)

File Edit Format Run Options Window Help

```
marks = 0
try:
    marks = int(input("Type term test marks : "))
    if marks<0 or marks>100:
        raise Exception ("Value must be in the range of 0 to 100")
    else:
        print("Marks with in the range")

except ValueError:
    print("Not a value")

except Exception:
    print("Marks not in range")

else:
    print("Program worked successfully")
```

Program Output

```
========== RESTART: D:\Nishan\Python\SimpleProgram2Raise.py ===
Type term test marks : 20
Marks with in the range
Program worked successfully
>>>
========== RESTART: D:\Nishan\Python\SimpleProgram2Raise.py ===
Type term test marks : 102
Marks not in range
>>>
========== RESTART: D:\Nishan\Python\SimpleProgram2Raise.py ===
Type term test marks : abc
Not a value
>>>
```

Reference

- https://www.w3schools.com/python/python_try_except.asp
- https://docs.python.org/3/tutorial/errors.html

Any Questions?